CIS 635 Knowledge Discovery & Data Mining

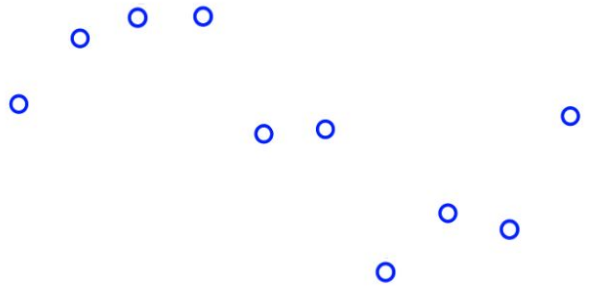
- Linear to Polynomial Regression
- Model Regularization

Plan

- LR to Polynomial Regression
- Regularization
  - Theory
  - Practical - Notebook presentation

Non linear data/function

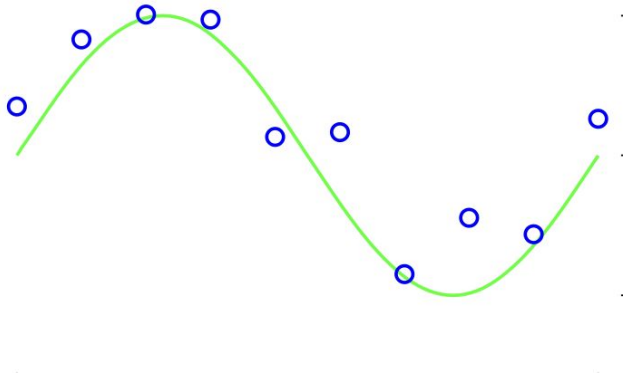
- Does this data points seem familiar matching a known function?



Non linear data/function

- Does this data points seem familiar matching a known function?
    - A Sinusoidal function

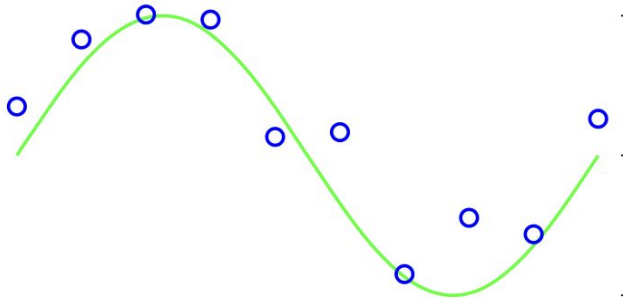
$$y(t) = A\sin(\omega t + \varphi) = A\sin(2\pi f t + \varphi)$$



Non linear data/function

- Does this data points seem familiar matching a known function?
    - A Sinusoidal function

$$y(t) = A\sin(\omega t + \varphi) = A\sin(2\pi f t + \varphi)$$

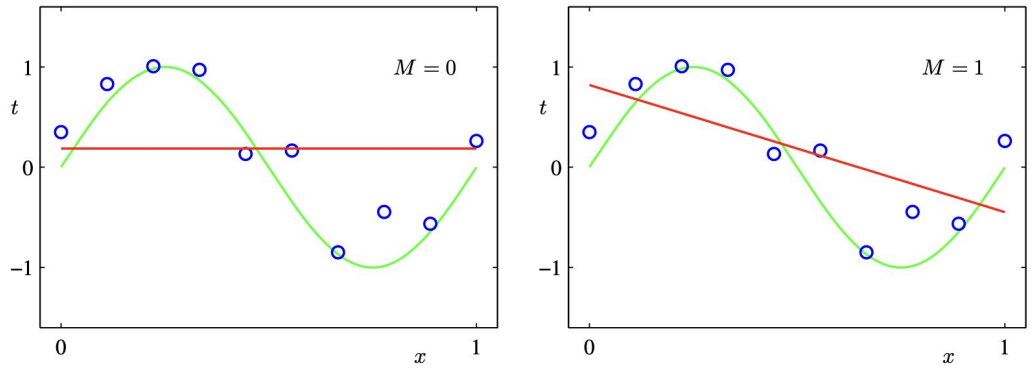


Clearly this is not a linear function; right?

Non linear data/function

- Does this data points seem familiar matching a known function?
- Can we approximate this function using LR ?

$$\hat{y} = \beta_0 + \beta_1 x$$



LR will not work; right?

What no-linear functions we are aware of?

- Can you recall any nonlinear function you learned at your high school/colleges?

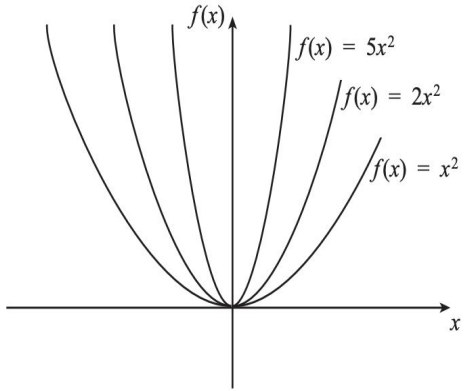
ref

What no-linear functions we are aware of?

- Can you recall any nonlinear function you learned at your high school/colleges?
- **Quadratic (x^2)**

$$f(x) = x^2, \qquad f(x) = 2x^2, \qquad f(x) = 5x^2.$$

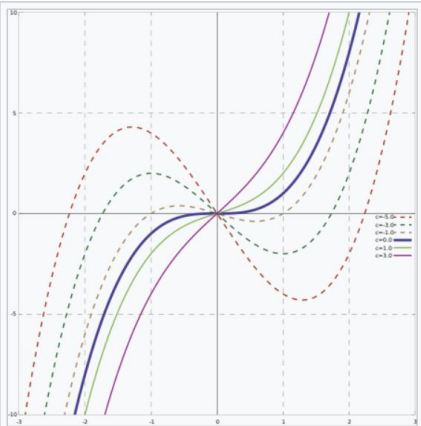
What is the impact of changing the coefficient of x^2 as we have done in these examples? One way to find out is to sketch the graphs of the functions.



ref

What no-linear functions we are aware of?

- Can you recall any nonlinear function you learned at your high school/colleges?
- **Cubic (x^3)**



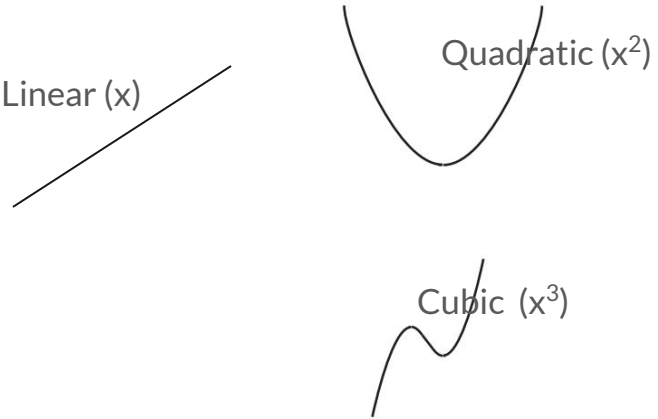
Cubic functions of the form $y = x^3 + cx$.
The graph of any cubic function is similar to such a curve.

ref

What no-linear functions we are aware of?

- Can you recall any nonlinear function you learned at your high school/colleges?
- Quadratic (x^2)
- Cubic (x^3)
-

$$\hat{y} = \beta_0 + \beta_1 x$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3 + \ldots$$

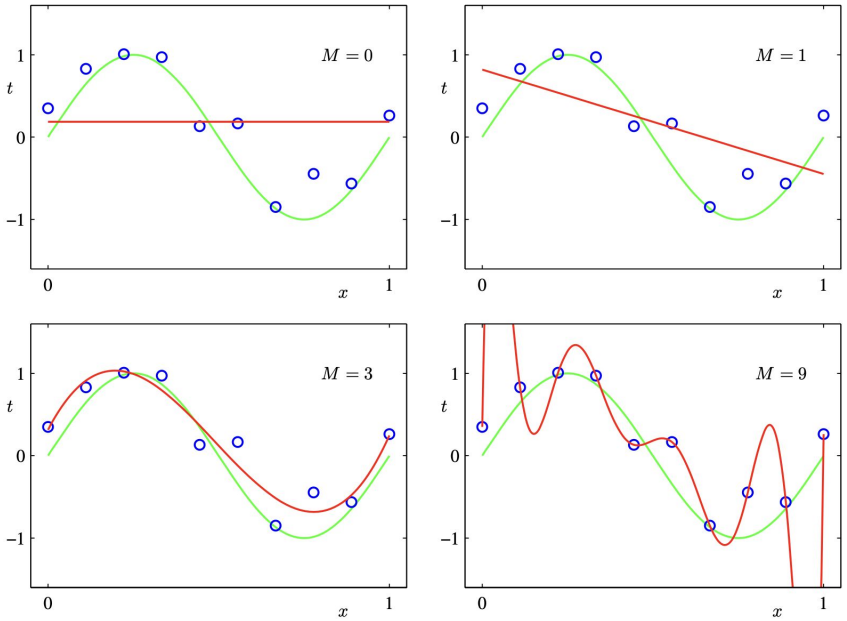


ref

LR to Polynomial Regression

- Polynomial function
  - **M is the order/degree of polynomial ..**

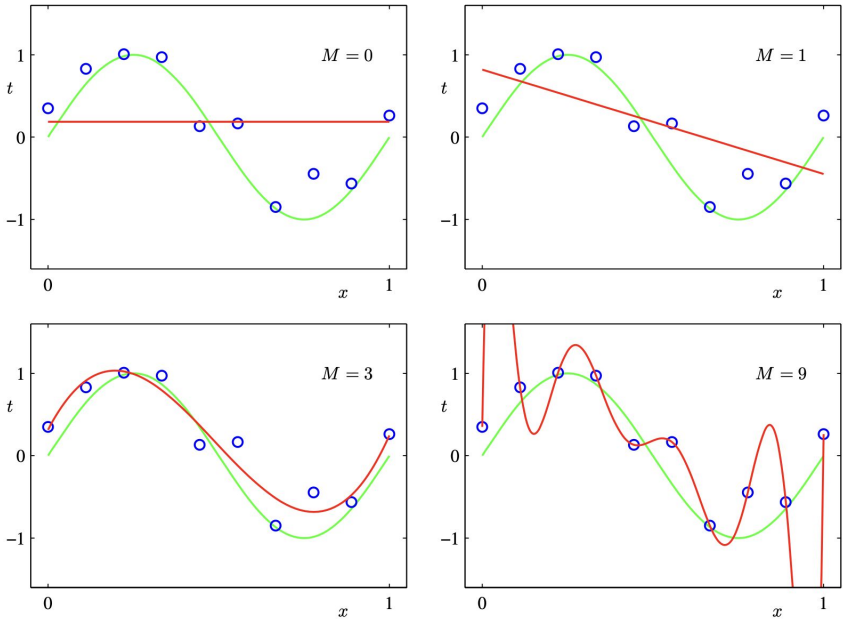
$$\hat{y} = \beta_0 + \beta_1 x$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3 + \ldots$$



LR to Polynomial Regression

- Polynomial function
  - *M* is the order/degree of polynomial ..
  - **Where to stop? What is the best *M*?**

$$\hat{y} = \beta_0 + \beta_1 x$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3 + \ldots$$

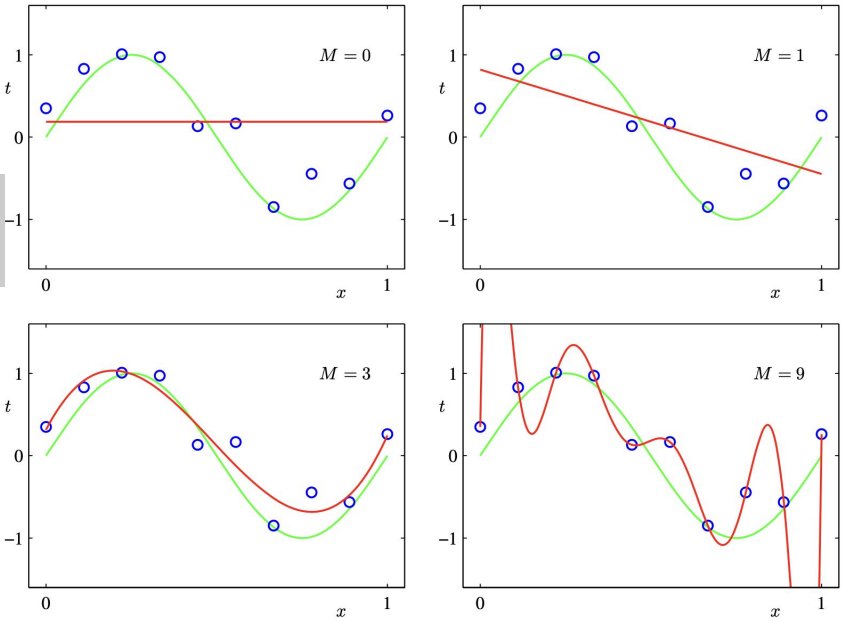


LR to Polynomial Regression

- Polynomial function
  - **M** is the order/degree of polynomial ..
  - **Where to stop? What is the best M?**

Good news is our gradient descent (iterative learning) remains the same!

$$\hat{y} = \beta_0 + \beta_1 x$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3 + \ldots$$

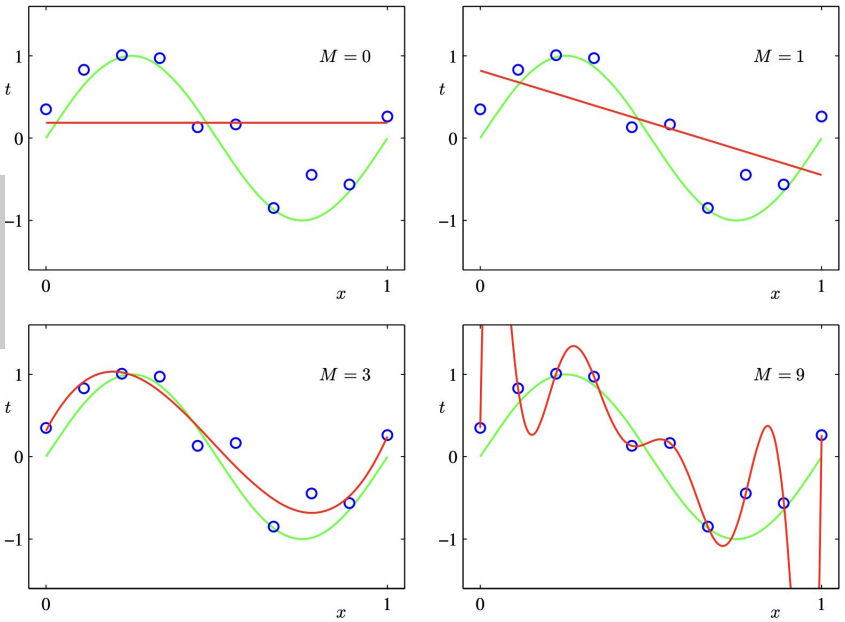


LR to Polynomial Regression

- Polynomial function
  - M is the order ..
  - **Where to stop? What is the best M?**

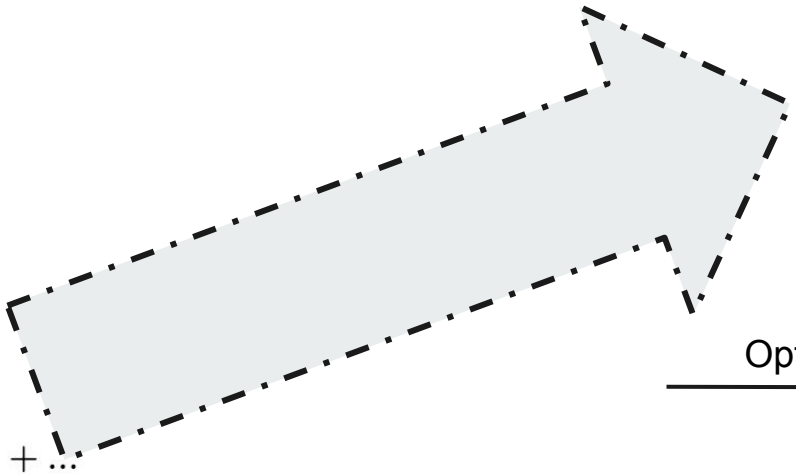
- Good news is our gradient descent (iterative learning) remains the same!
- You only need to change your objective function (from LR to Polynomial LR)

$$\hat{y} = \beta_0 + \beta_1 x$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3 + \ldots$$



LR to Polynomial Regression

$$\hat{y} = \beta_0 + \beta_1 x$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3 + \dots$$



Model

$$\hat{y} = \beta_0 + \beta_1 x$$

$$\Theta = \{\beta_0, \beta_1\}$$

$$\epsilon = |\hat{y} - y|$$

Optimization function

$$E_\Theta = \frac{1}{2} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2$$

$$\Theta^* = \text{argmin}_\Theta \, E\{(x_i, y_i)\}_{i=1,\cdots,N}$$

Our model got a little bigger: 2 params to *M* param



GPT

I know one of your tricks; get you soon!!



Our model yesterday

Our model got a little bigger: 2 params to *M* param



GPT

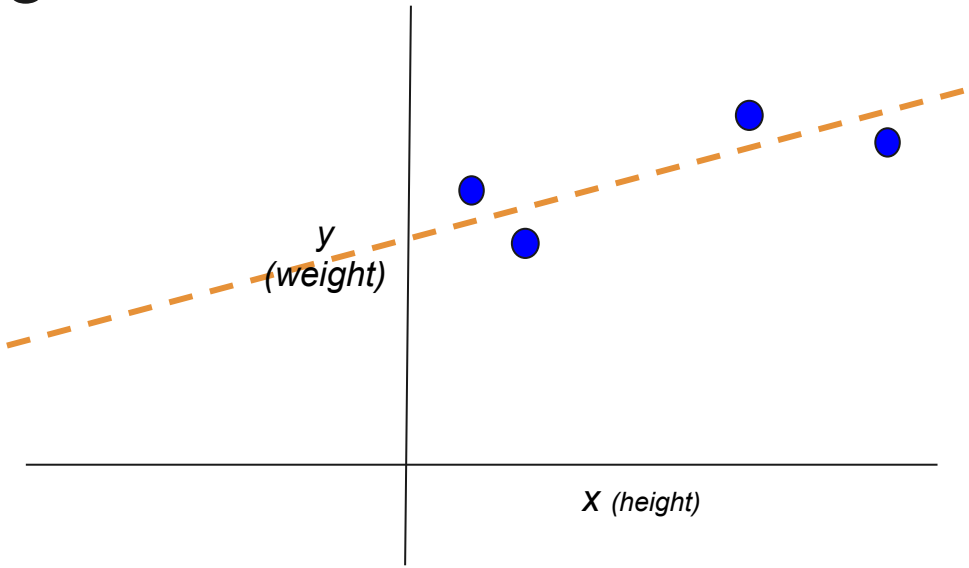


Our model today



Regularization

Regularization



So, essentially we are fitting a function; right?

Model

$$\hat{y} = \beta_0 + \beta_1 x$$

$$\Theta = \{\beta_0, \beta_1\}$$

Fitting Error

$$\epsilon = |\hat{y} - y|$$

Optimization function

$$E_\Theta = \frac{1}{2}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2$$

$$\Theta^* = \text{argmin}_\Theta \, E\{(x_i, y_i)\}_{i=1,\cdots,N}$$

Regularization

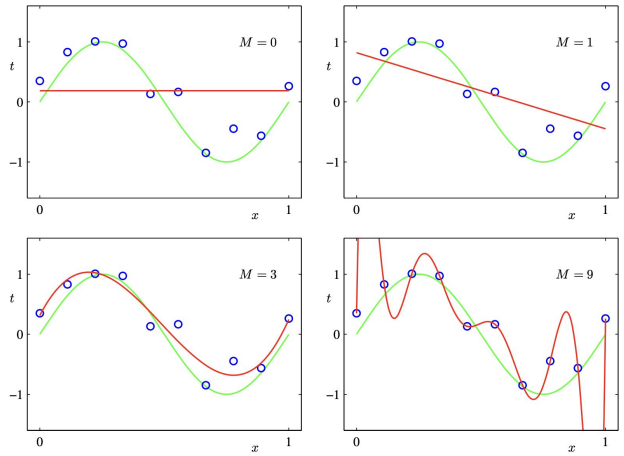


Table 1.1 Table of the coefficients $\mathbf{w}^*$ for polynomials of various order. Observe how the typical magnitude of the coefficients increases dramatically as the order of the polynomial increases.

	$M=0$	$M=1$	$M=6$	$M=9$
$ß_0$	0.19	0.82	0.31	0.35
$ß_1$		-1.27	7.99	232.37
$ß_2$			-25.43	-5321.83
$ß_3$			17.37	48568.31
$ß_4$				-231639.30
$ß_5$				640042.26
$ß_6$				-1061800.52
$ß_7$				1042400.18
$ß_8$				-557682.99
$ß_9$				125201.43

Regularization

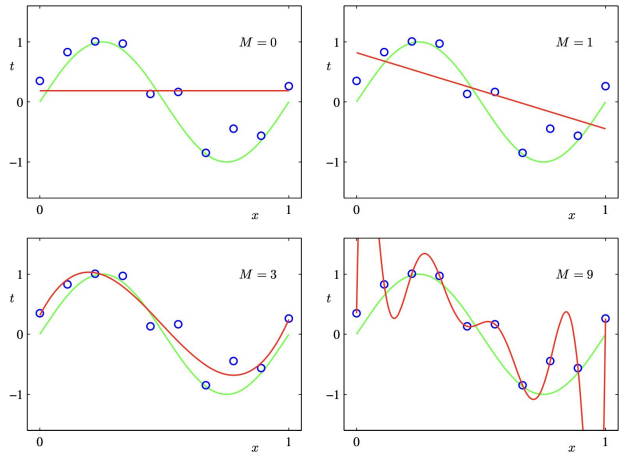
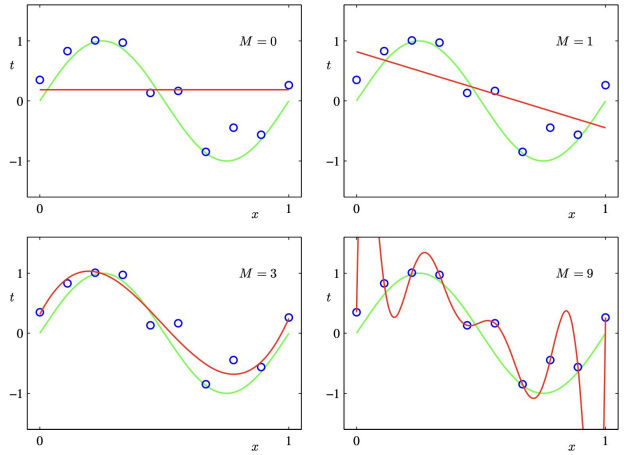


Table 1.1 Table of the coefficients $\mathbf{w}^*$ for polynomials of various order. Observe how the typical magnitude of the coefficients increases dramatically as the order of the polynomial increases.

	$M=0$	$M=1$	$M=6$	$M=9$
$ß_0$	0.19	0.82	0.31	0.35
$ß_1$		-1.27	7.99	232.37
$ß_2$			-25.43	-5321.83
$ß_3$			17.37	48568.31
$ß_4$				-231639.30
$ß_5$				640042.26
$ß_6$				-1061800.52
$ß_7$				1042400.18
$ß_8$				-557682.99
$ß_9$				125201.43

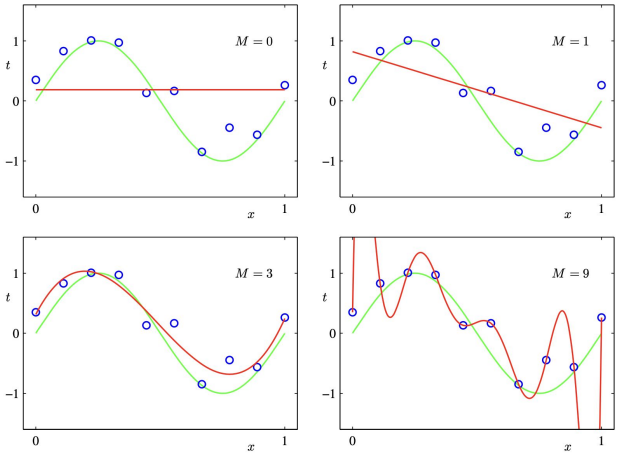
Absolute values are increasing

Regularization



$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2$$

Regularization



$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2$$

Regularizer

$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

Regularization

$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2$$

Regularizer

$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

Table 1.1 Table of the coefficients $\mathbf{w}^\star$ for polynomials of various order. Observe how the typical magnitude of the coefficients increases dramatically as the order of the polynomial increases.

	$M=0$	$M=1$	$M=6$	$M=9$
β_0	0.19	0.82	0.31	0.35
β_1		-1.27	7.99	232.37
β_2			-25.43	-5321.83
β_3			17.37	48568.31
β_4				-231639.30
β_5				640042.26
β_6				-1061800.52
β_7				1042400.18
β_8				-557682.99
β_9				125201.43

Abs values
Are increasing

Regularization

$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2$$



$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

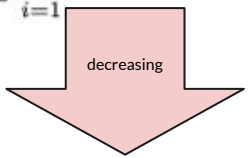


Table 1.1 Table of the coefficients $\mathbf{w}^\star$ for polynomials of various order. Observe how the typical magnitude of the coefficients increases dramatically as the order of the polynomial increases.

	$M=0$	$M=1$	$M=6$	$M=9$
β_0	0.19	0.82	0.31	0.35
β_1		-1.27	7.99	232.37
β_2			-25.43	-5321.83
β_3			17.37	48568.31
β_4				-231639.30
β_5				640042.26
β_6				-1061800.52
β_7				1042400.18
β_8				-557682.99
β_9				125201.43

Abs values
Are increasing

Regularization

$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2$$



$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

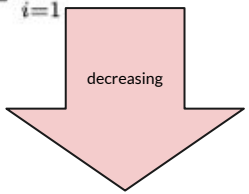


Table 1.1 Table of the coefficients $\mathbf{w}^\star$ for polynomials of various order. Observe how the typical magnitude of the coefficients increases dramatically as the order of the polynomial increases.

	$M=0$	$M=1$	$M=6$	$M=9$
β_0	0.19	0.82	0.31	0.35
β_1		-1.27	7.99	232.37
β_2			-25.43	-5321.83
β_3			17.37	48568.31
β_4				-231639.30
β_5				640042.26
β_6				-1061800.52
β_7				1042400.18
β_8				-557682.99
β_9				125201.43

Abs values
Are increasing

Regularization

$$\tilde{E}(\Theta) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2$$



$$\tilde{E}(\Theta) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

decreasing

How to control this?

Table 1.1 Table of the coefficients $\mathbf{w}^\star$ for polynomials of various order. Observe how the typical magnitude of the coefficients increases dramatically as the order of the polynomial increases.

	$M=0$	$M=1$	$M=6$	$M=9$
β_0	0.19	0.82	0.31	0.35
β_1		-1.27	7.99	232.37
β_2			-25.43	-5321.83
β_3			17.37	48568.31
β_4				-231639.30
β_5				640042.26
β_6				-1061800.52
β_7				1042400.18
β_8				-557682.99
β_9				125201.43

Abs values
Are increasing

Regularization

$$\tilde{E}(\Theta) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2$$



$$\tilde{E}(\Theta) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

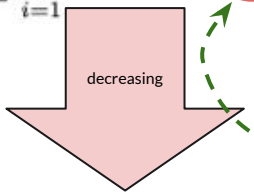


Table 1.1 Table of the coefficients $\mathbf{w}^*$ for polynomials of various order. Observe how the typical magnitude of the coefficients increases dramatically as the order of the polynomial increases.

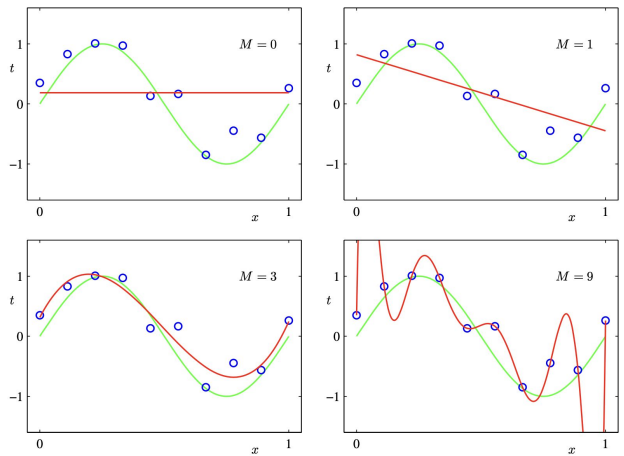
	$M=0$	$M=1$	$M=6$	$M=9$
β_0	0.19	0.82	0.31	0.35
β_1		-1.27	7.99	232.37
β_2			-25.43	-5321.83
β_3			17.37	48568.31
β_4				-231639.30
β_5				640042.26
β_6				-1061800.52
β_7				1042400.18
β_8				-557682.99
β_9				125201.43

Abs values
Are increasing

How to control this?

Lamda is called the
Hyper Parameter of this model

Polynomial Regression with Regularization

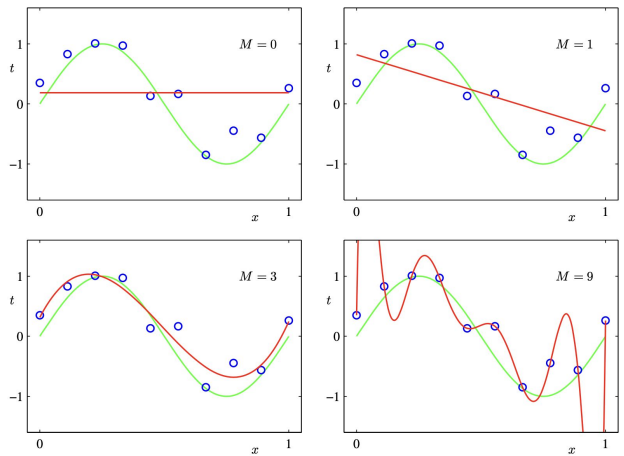


Learned function is **nonlinear**

$$\hat{y} = \beta_0$$
$$\hat{y} = \beta_0 + \beta_1 x$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3 + \ldots$$

$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

Polynomial Regression with Regularization



$$\hat{y} = \beta_0$$
$$\hat{y} = \beta_0 + \beta_1 x$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3$$
$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_2 x^3 + \ldots$$

$$\tilde{E}(\mathbf{\Theta}) = \frac{1}{2}\sum_{i=1}^{N}\{\hat{y}_i - y_i\}^2 + \lambda\sum_{j=1}^{p}\beta_j^2$$

Learned function is **nonlinear**

Model (still) **linear**

Classification

- General Idea (two steps process)
  - LR (Bias Only)
  - LR (general)

Notebook presentation

- Without regularizer
- With regularizer

Predictive modeling: Regression (diabetes)

Predictive modeling: Classification

Delivery Plan

Introduction (KDD & Python) and networking (2W)

EDA (2W)

Supervised Modeling (4W)

Unsupervised Learning (2W)

Sequence Analysis and Modeling (1W)

Special Topics: DL, NN

Project (Learning by Doing)

Week	Week of	Topics
1	Aug. 31	Introduction to Knowledge Discovery & Data Mining; Python Basics
2	Sept. 7	Python for Data Science; Descriptive Statistics & Data Visualization
3	Sept. 14	Exploratory Data Analysis (EDA) I
4	Sept. 21	Exploratory Data Analysis (EDA) II & Case Studies
5	Sept. 28	Similarity & Distance Measures; K-Nearest Neighbors
6	Oct. 5	Predictive Modeling: Regression
7	Oct. 12	Predictive Modeling: Classification I
8	Oct. 19	Predictive Modeling: Classification II & Model Evaluation
9	Oct. 26	*Fall Break*
10	Nov. 2	Unsupervised Learning: Clustering
11	Nov. 9	Clustering & Dimensionality Reduction
12	Nov. 16	Sequence Analysis & Modeling
13	Nov. 23	Special Topics: Deep Learning & Neural Networks
14	Nov. 30	Special Topics: Emerging Data-Mining Techniques
15	Dec. 7	Final Project Presentations & Course Wrap-Up
16	Dec. 14	Final Examination & Evaluation Week

Delivery Plan

Introduction (KDD & Python) and networking (2W)

EDA (2W)

Supervised Modeling (4W)

Unsupervised Learning (2W)

Sequence Analysis and Modeling (1W)

Special Topics: DL, NN

Project (Learning by Doing)

Week	Week of	Topics
1	Aug. 31	Introduction to Knowledge Discovery & Data Mining; Python Basics
2	Sept. 7	Python for Data Science; Descriptive Statistics & Data Visualization
3	Sept. 14	Exploratory Data Analysis (EDA) I
4	Sept. 21	Exploratory Data Analysis (EDA) II & Case Studies
5	Sept. 28	Similarity & Distance Measures; K-Nearest Neighbors
6	Oct. 5	Predictive Modeling: Regression
7	Oct. 12	Predictive Modeling: Classification I
8	Oct. 19	Predictive Modeling: Classification II & Model Evaluation
9	Oct. 26	*Fall Break*
10	Nov. 2	Unsupervised Learning: Clustering
11	Nov. 9	Clustering & Dimensionality Reduction
12	Nov. 16	Sequence Analysis & Modeling
13	Nov. 23	Special Topics: Deep Learning & Neural Networks
14	Nov. 30	Special Topics: Emerging Data-Mining Techniques
15	Dec. 7	Final Project Presentations & Course Wrap-Up
16	Dec. 14	Final Examination & Evaluation Week